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## BIOLOGY

### Paper 3: Applications paper and Planning Question

9648/03

29<sup>th</sup> August 2014

2 hours

Additional Materials: Writing Paper

### READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

#### Section A]

Answer all questions

#### Section B]

Answer all questions. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [ ] at the end of each question or part of the question.

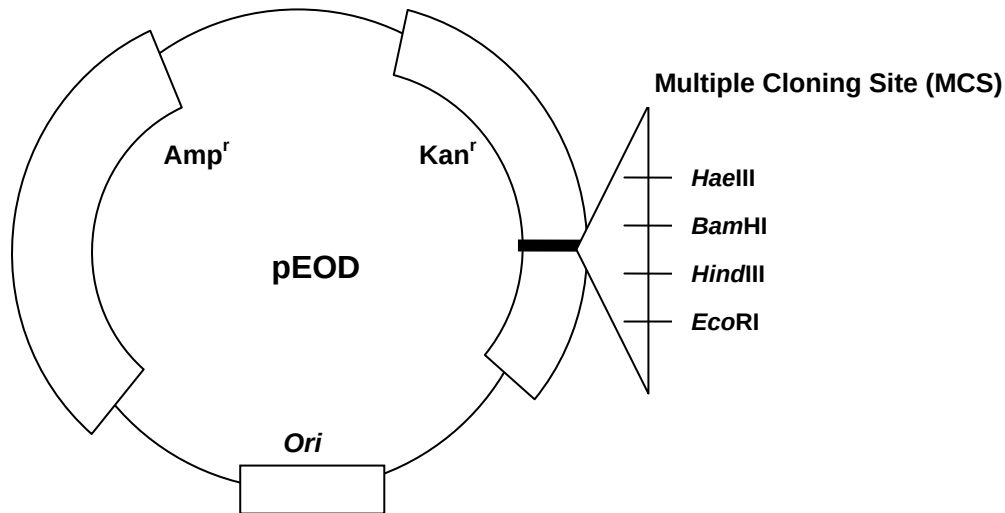
| For Examiner's Use |    |
|--------------------|----|
| Section A          | 52 |
| 1 [12]             |    |
| 2 [14]             |    |
| 3 [14]             |    |
| 4 [12]             |    |
| Section B          | 20 |
| 5a [6]             |    |
| 5b [6]             |    |
| 5c [8]             |    |
| TOTAL              | 72 |

## Section A

Answer **all** questions in this section.

1. **Fig. 1.1** shows the plasmid pEOD.

Plasmid pEOD is a cloning vector used in creating recombinant DNA. It contains two selectable marker genes, Kan<sup>R</sup> (kanamycin resistance gene) and Amp<sup>R</sup> (ampicillin resistance gene). A multiple cloning site (MCS) is located within the Kan<sup>R</sup> gene.



**Fig. 1.1**

A target sequence of human DNA is replicated using the Polymerase Chain Reaction (PCR).

The target DNA sequence is shown in **Fig. 1.2**.

5' - GGATCCATCCCGATCGA TCC T TT AAGCTAGCTAGGATCC - 3'  
3' - CCTAGGTAAGGGCTAGCT AGGAAA T TCGATCGATCCTAGG - 5'

**Fig. 1.2**

(a) Design two 8-base long primers for the sequence to be replicated.

[2]

Primer 1: 5' – GGATCCAT – 3'

Primer 2: 3' – ATCCTAGG – 5'

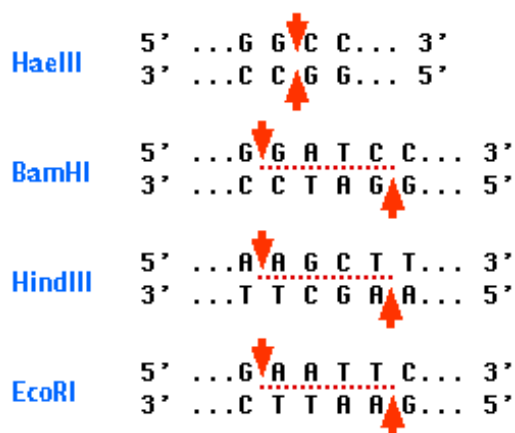
- 5' and 3' indicated correctly on both primers- 1m
- correct sequence of both primers – 1m
- 1m awarded if one of the primer has correct sequence and direction indicated.

(b) State three differences between the process of PCR and natural DNA replication in cells. [3]

| Basis of comparison)         | Differences                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Nature of primers         | DNA replication involves RNA primers while PCR requires DNA primers.                                                                                         |
| 2. Location                  | DNA replication takes place in the nucleus of the cell while PCR is automated/takes place in a thermocycler.                                                 |
| 3. Enzymes involved          | DNA replication involves DNA polymerase III and DNA polymerase I while PCR involves the enzyme <i>Taq</i> polymerase.                                        |
| 4. Proof-reading             | In DNA replication, the daughter strands are proofread by DNA polymerase I but there is no proofreading of daughter strands in PCR by <i>Taq</i> polymerase. |
| 5. Unzipping of template DNA | In PCR, high temperatures are required for denaturing the strands while in DNA replications, the enzyme helicase unwinds the strands.                        |
| 6. Synthesis of primers      | In DNA replication, primase synthesises the RNA primers, but in PCR, the primers are added in/primase is not involved.                                       |

After the target DNA has been subjected to PCR, it is cleaved using restriction enzymes and then inserted into pEOD. **Fig. 1.3** shows the types of restriction enzymes that can be used to cleave pEOD at the Multiple Cloning Site (MCS) shown in **Fig. 1.1**

**Restriction enzyme      Specific target base sequence of DNA**



**Fig. 1.3**

(c)(i) State the restriction enzyme used to cleave the target DNA sequence before it is inserted into vector pEOD. [1]

- *BamHI*

(ii) Outline the steps that must subsequently be followed to produce a recombinant plasmid. [3]

1. The target sequence and plasmid are cleaved using *Bam*HI/ same restriction enzyme to give sticky ends/ staggered ends.
2. The target sequence and the plasmid are then mixed together to allow sticky ends/ staggered ends to anneal by hydrogen bonding / complementary base pairing.
3. DNA ligase is added to form phosphodiester bonds/seal the nicks between the target sequence and the plasmid.

(d) Describe and explain how bacteria transformed with this recombinant plasmid are identified using selective agar media. L2 [3]

1. The bacteria are first plated onto nutrient agar plates containing the ampicillin, this would select for bacteria colonies transformed with plasmid/pEOD (recombinant & reannealed plasmids)
2. Replica-plate the bacteria colonies grown on the ampicillin plate onto nutrient agar plates containing kanamycin
3. Obtain bacteria colonies able to grow on ampicillin plate but unable to grow on kanamycin plate as bacteria transformed with recombinant plasmid / OWTTE
4. Bacteria transformed with recombinant plasmid will be unable to grow on kanamycin plate due to insertional inactivation of the Kan<sup>R</sup> marker gene

[Total: 12]

- 2 In Singapore, more than 20% of all cancers diagnosed in women are breast cancers. Mutation in BRCA-1 gene is strongly associated with the heritable form of this disease. BRCA-1 gene codes for a protein involved in cell cycle regulation and DNA repair. Fig 2.1 shows the pedigree and RFLP profile of a family with history of heritable breast cancer.

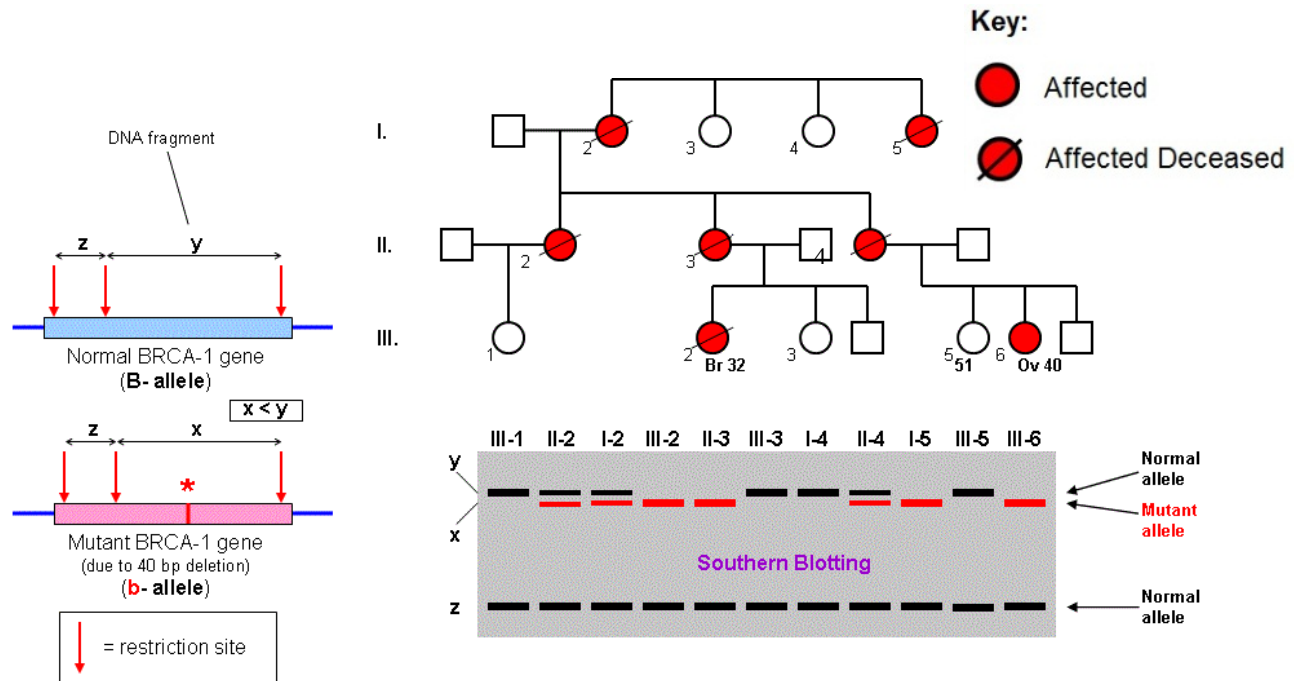


Fig 2.1

- (a) State the genotype(s) that will lead to development of this form of breast cancer among females. [1]
- Bb and bb
- (b) Explain how the mutation on b-allele allows for the RFLP analysis of this heritable form of breast cancer. [3]
1. The 40 bp deletion on b-allele resulted in a shorter fragment x as compared to fragment y of the normal B-allele
  2. The difference in size of fragment X and Y allows for separation by gel electrophoresis
  3. visualized by radioactively-labelled / fluorescent-labelled hybridized probe for RFLP analysis

**(c)** Account for the band patterns of the following female individuals: [4]

1. II-2 is a deceased female with one copy of b-allele and B-allele / heterozygous Bb individual / OWTTE
2. Restriction enzyme digestion results in three bands (fragments x, y and z) due to deletion mutation on 1 copy of b-allele.
3. III-6 is an affected female homozygous for b-allele
4. Hence there will only be two bands (fragment x and z) due to absence of normal B-allele / OWTTE.

**(d)** Explain how bands patterns are formed in RFLP. [3]

1. The agarose gel is a cross-linked matrix and functions as a 'molecular sieve' where the matrix forms little pores (holes) through which DNA can travel through.
2. negatively charged DNA molecules migrate toward the positive end of the field/anode.
3. DNA fragments separated according to their size; The smaller molecules move faster than the larger one, so they migrate further than the larger molecules / Vice Versa, forming the band patterns.
4. Southern Blotting where DNA fragments are transferred to the nitrocellulose membrane
5. Incubate with radioactively/fluorescent labelled probe and visualize it using autoradiography.

[any 3 of the above]

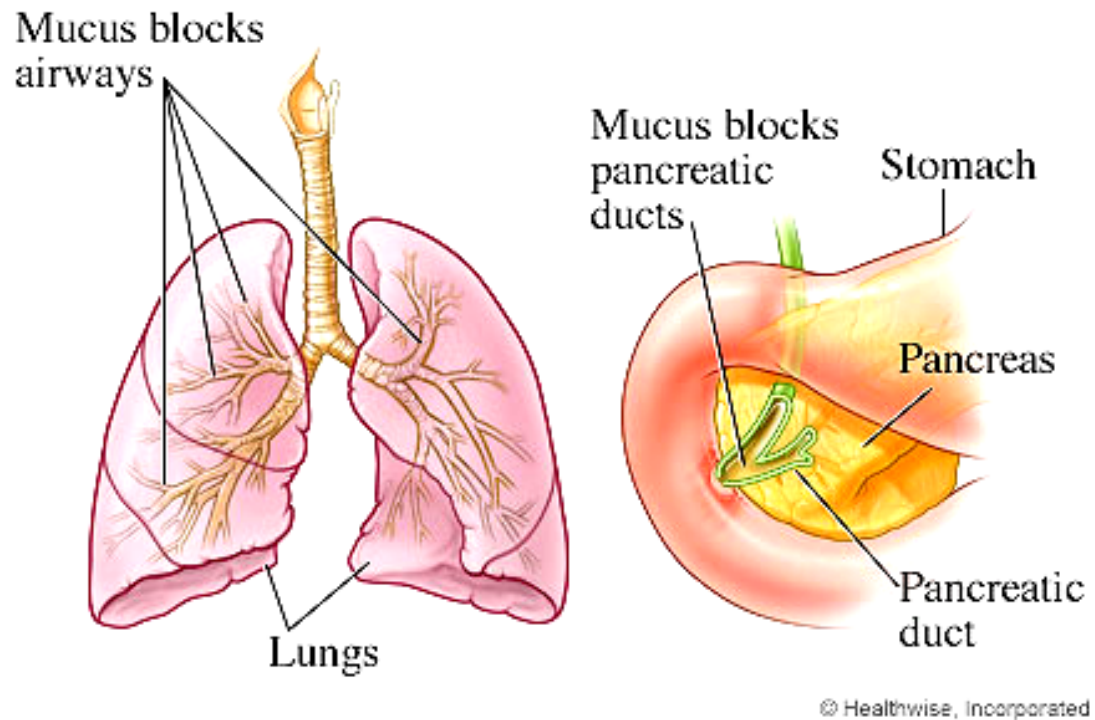
**(e)** Suggest how the probe used in the RFLP analysis can be produced. [3]

1. Obtain BRCA-1 gene sequence using bioinformatics / gene library / genetic database as template.
2. Synthesize short complementary sequences that target the polymorphic region between fragment Z and X/Y of disease and normal allele.
3. Add radioactive / fluorescence label to the probe.

**[Total:14]**

- 3 The cystic fibrosis transmembrane conductance regulator (CFTR) gene encodes for the CFTR chloride channel. The CFTR chloride channel is responsible for transporting chloride ions in and out of the cell.

Individuals who suffer from cystic fibrosis (CF) have genetic mutations on the CFTR gene which results in dysregulation of chloride ions across the cell. As shown in Fig. 3.1, thick mucus produced in CF patients clogs up the airways of the lungs and the branches of the pancreatic duct.



**Fig. 3.1**

**(a)** State two more organs not shown in Fig. 3.1 which may be affected by the cystic fibrosis disease. [2]

- Liver bile duct / fallopian tubes/vas deferens / small intestines / sweat glands

**(b)** Explain how a named mutation of the CFTR gene can lead to the condition shown in Fig. 3.1. [4]

1. A common genetic mutation of cystic fibrosis is the deletion of 3 base pairs at codon position 508 in the CFTR mRNA sequence.
2. As a result, the amino acid phenylalanine is missing at position 508 in the protein.
3. The mutant protein does not leave the endoplasmic reticulum for further processing in the Golgi apparatus, hence there is no functional CFTR present on the cell surface membrane.
4. Normal efflux of  $\text{Cl}^-$  out from the cells is prevented &  $\text{Cl}^-$  ions accumulate in the cells
5. Being negatively charged, the build-up of  $\text{Cl}^-$  ions in the cells causes the influx of  $\text{Na}^+$  ions into the cells to balance the charges.
6. High concentration of ions inside the cell in turn prevents water from leaving the cells causing mucus produced to be thicker and more viscous.

**(c)** With the advancement of medical science, cystic fibrosis can now be treated via gene therapy methods.

(i) Describe one viral method of gene therapy treatment for cystic fibrosis. [2]

1. Modified adenovirus harbouring the normal CFTR gene can be used to treat cystic fibrosis.
2. In which, the modified adenovirus will infect patient's affected cells and deliver the normal gene into the patient's cells where it will express the normal CFTR protein.

(ii) Describe one non-viral method of gene therapy treatment for cystic fibrosis. [2]

1. Copies of the normal allele of the gene for cystic fibrosis are inserted into plasmid DNA which are then attached to liposomes in which positively-charged liposomes attract negatively-charged DNA to form complexes
2. The liposome-DNA complexes are then sprayed as an aerosol of fine droplets into the noses of recipient patients and enter cells through fusion with the cell surface membrane.

(d) (i) Suggest one advantage and one disadvantage of the viral method described in (b)(i)[2]

|              |                                                                                                                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADVANTAGE    | 1. Modified adenovirus does not integrate into host cell genome and hence will not cause disruption of the integrity of certain genes which may lead to cancer.                                                          |
| ADVANTAGE    | 2. Modified adenovirus can infect both dividing and non-dividing cells.                                                                                                                                                  |
| ADVANTAGE    | 3. There is no integration into host cell genome. Hence there is no disruption of genes.                                                                                                                                 |
| DISADVANTAGE | 4. As adenovirus does not insert its DNA into the host DNA, thus if the infected host cell divides, the therapeutic DNA is not replicated at the same time. Treatment is therefore short-lived and needs to be repeated. |
| DISADVANTAGE | 5. Lipofection has a low transfection efficiency and therefore this method is inefficient.                                                                                                                               |

(Max 2: must have one advantage and one disadvantage)

- (ii) Suggest one advantage and one disadvantage of the non-viral method described in (b)(ii) [2]

|              |                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------|
| ADVANTAGE    | 1. Usage of liposome complexes is non-viral method which will not cause an immune response in patients. |
| ADVANTAGE    | 2. Liposome complexes are easy to prepare without the need to handle viruses.                           |
| DISADVANTAGE | 3. The effect of the treatment can only last about a week and repeated treatments are required.         |

(Max 2: must have one advantage and one disadvantage)

[Total: 14]

#### 4 Planning Question

A mutant strain of *Escherichia coli* (*E.coli*) has a disabled lac operon. The enzyme  $\beta$ -galactosidase, which is encoded by *lacZ* of the operon, has part of the polypeptide chain missing and is inactive. This mutant strain is referred to as **lac<sup>-</sup>**.

In genetic engineering, a modified F plasmid can be used as a cloning vector. This modified plasmid has part of the lac operon (*lacZ* gene) and is able to code for functional  $\beta$ -galactosidase. An active enzyme is synthesized if the plasmid is transferred to the **lac<sup>-</sup>** strain of *E.coli*.

Fig. 4.1 shows a simplified diagram of this plasmid.

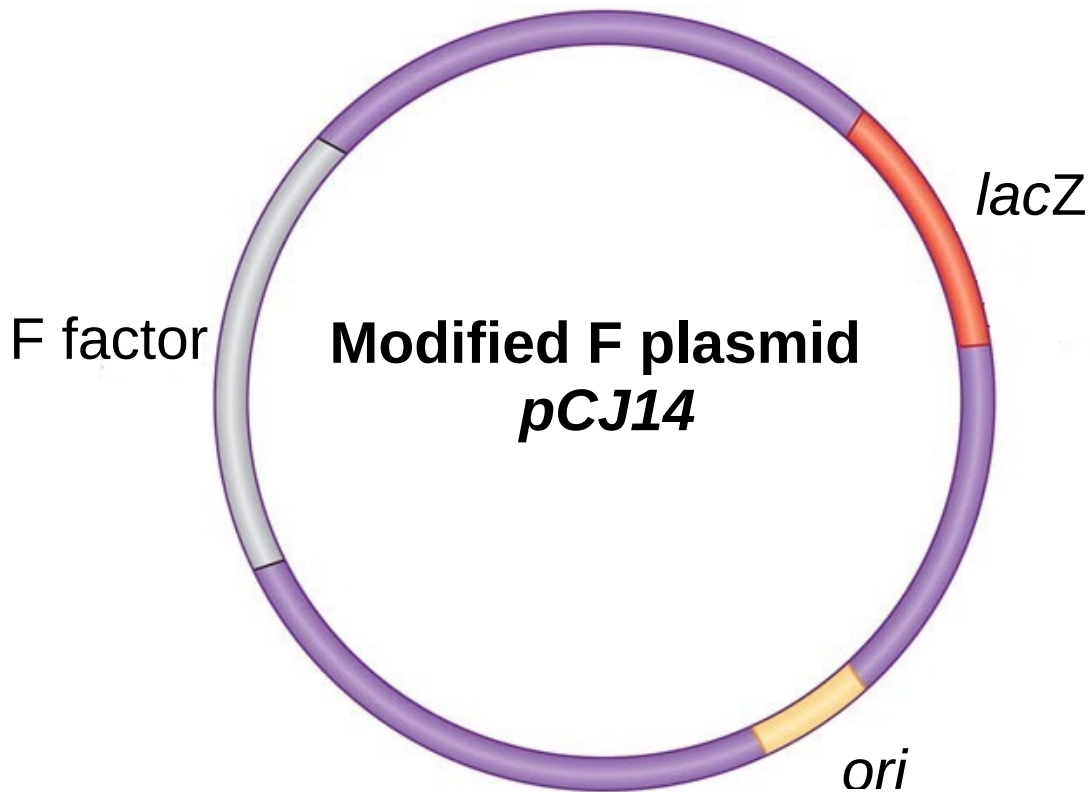


Fig. 4.1

The presence of the active enzyme  $\beta$ -galactosidase in colonies of **lac<sup>-</sup>** *E.coli* containing the modified F plasmid (**F<sup>+</sup>**) can be detected using nutrient agar containing the substrate, X-gal. The active enzyme hydrolyses X-gal to form a blue coloured compound. Without this plasmid the **lac<sup>-</sup>** colonies are white.

You are required to plan an experiment to investigate if the **efficiency of conjugation is dependent on the duration of mating time**. The transfer of the modified F plasmid into the **lac<sup>-</sup>** strain of *E. coli* from a **his<sup>-</sup>** strain of *E. coli* containing the modified plasmid (**F<sup>+</sup>**). The **his<sup>-</sup>** strain of *E. coli* lacks the genes to synthesise the amino acid histidine. It can therefore grow **only** on nutrient media to which histidine has been added.

At least 15 minutes of mating time is required for conjugation.

**Conjugation efficiency** is calculated as follows:

**(Average number of blue colonies/Average number of blue and white colonies) x 100%**

**Warning: X-gal growth medium is an irritant to eyes, skin and the respiratory system.**

Your planning must be based on the assumption that you have been provided with the following equipment and materials:

- Two cultures of the bacterium *Escherichia coli* (*E.coli*) are provided:

**Culture A (8 ml):** F<sup>-</sup>, lac<sup>-</sup>, his<sup>+</sup> *E.coli*,

**Culture B (8 ml):** F<sup>+</sup>, lac<sup>+</sup>, his<sup>-</sup> *E.coli*.

- Growth medium (agar) without the amino acid histidine, X-gal provided separately
- Petri dishes
- Boiling tubes
- Sterile cotton wool plug (for boiling tube)
- Sterile graduated Pasteur pipettes
- Stopwatch
- Sterile spreader
- Disinfectant solution
- Bunsen flame and lighter
- Incubator set at 37°C
- Masking tapes
- Disposable gloves
- Protective goggles

Your plan should have a clear and helpful structure to include:

- A description of the method used including the scientific reasoning behind the method,
- An explanation of the dependent and independent variables involved, relevant, clearly labeled diagrams,
- How you will record your results and ensure they are as accurate and reliable as possible,
- Proposed layout of results tables and graphs with clear headings and labels,
- The correct use of technical and scientific terms,
- Relevant risks and precautions taken

**[Total: 12]**

## Summary of Mark Scheme

| Marking Points                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1. Conjugation involves transfer of genetic material between bacteria cells via sex-pilus hence it would be time dependent. Longer mating time will lead to higher conjugation efficiency.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1    |
| 2. Conjugation occurs when there is transfer of genetic material (DNA) via sex pilus.<br>3. <i>LacZ</i> gene in the modified F plasmid serves as a genetic marker which is identifiable after conjugation.<br>4. $\beta$ -galactosidase encoded by the <i>lacZ</i> gene is an enzyme capable of breaking down colourless Xgal into a blue coloured compound.<br>5. Colonies of Culture A will appear blue if they have successfully undergone conjugation, otherwise they will remain white.<br>6. Since conjugation is a time-dependent process, the mixture between donor (culture B) and recipient (culture A) cells could be transferred for plating on culture media at fixed time intervals (i.e. every 5 mins).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2    |
| 7. <u>Independent Variable</u> :<br>Mating time between mixture of culture A and B (15, 20, 25, 30 and 35 mins)<br>8. <u>Dependent Variables</u> :<br>Conjugation efficiency is calculated as follows:<br>(Average number of blue colonies/Average number of blue and white colonies) x 100%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1    |
| Aseptic Techniques<br>9. Procedure indicate aseptic environment to prevent contamination (e.g. cleaning workspace with disinfectant / working within perimeter of lighted Bunsen flame / use of sterile apparatus and media)<br><br>Preparing growth media plates for <i>E.coli</i> cells<br>10. Autoclave the growth media (121 ° C, 30 minutes) and wait for the solution to cool (should be cool enough to handle but not cold) and pour carefully into the Petri dishes avoiding bubbles as much as possible. Addition of X-gal and allow the growth media to solidify into agar.<br>11. Prepare growth media (agar) with X-gal without histidine (16 plates, 3 plates for each time interval + 1 negative control)<br><br>Conjugation process (Mixing)<br>12. Label 3 boiling tubes as follows: <ul style="list-style-type: none"> <li>• Tube A – Culture A</li> <li>• Tube B – Culture B</li> <li>• Tube C – Mixture of Culture A and B</li> </ul> 13. Using separate sterile pipettes, transfer 5 ml of <i>E.coli</i> culture A and 5ml of <i>E.coli</i> culture B into Tube C. Ensure sterile techniques are used to avoid contamination.<br>14. Mix well the bacterial mixture in Tube C.<br>15. Allow the mixture in Tube C to stand undisturbed for 15 mins.<br><br>Plating <i>E.coli</i> cells<br>16. Resuspend mixture before plating.<br>17. Remove 1ml of the mixture in Tube C after 15 minutes and spread the <i>E.coli</i> culture over the entire surface of the agar in one of the plates using a spreader. Seal plate with masking tape and incubate them in a 37°C overnight<br>18. Repeat step 12 to 16 at each time interval (i.e. 20, 25, 30 and 35 minutes).<br><br>Replicates<br>19. Repeat entire experiment (Step 11 – Step 17) twice to obtain another two sets of observations. | 4    |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p>Identification of <i>E.coli</i> cells (Culture A) that has undergone conjugation</p> <p>20. Removes plates from the incubator. Count the number of white and blue colonies and record in the table of results.</p> <p>21. Controlled Variables (any one of the following)</p> <ul style="list-style-type: none"> <li>• Same volume of culture A and B in the mixture</li> <li>• Same volume of mixture plated onto each petri dish with X-gal growth media for each time interval.</li> <li>• Constant temperature maintained by water bath / incubator</li> <li>• Time for bacterial culture incubation (not conjugation time)</li> </ul> <p>Control</p> <p>22. Negative control: Same volume of bacterial culture A plated on X-gal growth media without mixing with culture B → all white colonies (since none of the colonies would be lac<sup>+</sup>)</p> <p>Recording observations</p> <p>23. Record the observations (See point 22)</p> <p>24. Conjugation efficiency is calculated as follows:<br/>(Average number of blue colonies/Average number of blue and white colonies) x 100%</p> |          |
| <p>25. *Risk: <i>E.coli</i> may cause disease.<br/>*Precaution: wear gloves and use antiseptic handwash to wash hands thoroughly after experiment<br/>Rationale: Practice aseptic techniques and proper handwash to prevent infection.</p> <p>26. *Risk: X-gal is an irritant.<br/>*Precaution: Wear gloves and eye protection when handling it.<br/>Rationale: Prevent irritation.</p> <p>27. *Risk: Lab may be contaminated with microorganism.<br/>*Precaution: Dispose all bacterial cultures and media properly in biohazard bags and clean work surfaces with alcohol<br/>Rationale: To prevent contamination of lab to prevent avenue for microorganisms to grow and protect oneself from accidental bacterial spillage</p>                                                                                                                                                                                                                                                                                                                                                                    | <b>1</b> |
| 28. Table of observation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>1</b> |
| 29. Graph                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>1</b> |
| <p>30. <i>Longer mating time will result in higher conjugation efficiency due to longer duration for transfer of genetic material via sex pilus.</i></p> <p>31. <i>E.coli</i> cells that have taken up F plasmids will express lacZ gene / β-galactosidase. This is due to transfer of genetic material via sex-pilus during conjugation, and thus be easily identified via culture on X-gal medium.</p> <p>32. <i>Blue colonies represent expression of lacZ gene / β-galactosidase as X-gal is broken down into blue compound.</i></p> <p>33. <i>Conjugation efficiency is calculated for culture A only as culture B would be unable to grow on culture media without histidine.</i></p> <p>34. Correct use of scientific terms/ correct scientific reasoning (Conjugation, Sex/F-pilus, Blue-white screening)</p>                                                                                                                                                                                                                                                                                 | <b>1</b> |

**[Total:12]**

## Section B

Answer **all** questions. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

**5 (a)** Outline the procedure for propagating plants by tissue culture technique. [6]

1. Explants such as shoot tip/root tip/small pieces of excised plant parts are taken from the leaf or stem / meristem are obtained from plant.
2. Surface of explants are sterilised using dilute sodium hypochlorite.
3. Explants are aseptically transferred to culture vessels containing nutrients and plant growth regulators
4. Cytokinin in the nutrient agar stimulate the cells of an explant to divide by mitosis forming a callus (undifferentiated, actively growing mass of cells)
5. The cells of the callus are induced by auxin and giberellin to differentiate into particular tissues e.g. roots & shoots. An organised tissue can develop into a plantlet.
6. When the plantlet is sufficiently large, it is transferred from the laboratory to the soil and grown out into a whole plant.

**(b)** Describe with an example the genetic engineering methodology which utilises bacteria that can improve the nutritional content of crops [6]

1. Two foreign genes, phytoene synthase (*Psy*) from daffodil and carotene desaturase (*CrtI1*) from *Erwinia uredovae* bacterium
2. are cloned into Ti plasmid.
3. These genes are enzymes that are involved in the synthesis of  $\beta$ -carotene, which will in turn be converted to Vitamin A.
4. The genes are placed after an endosperm-specific promoter
5. As the endosperm which will go on to develop into rice grains do not produce  $\beta$ -carotene and hence there will be no Vitamin present.
6. The Ti plasmid harbouring the foreign genes are transformed into *Agrobacterium tumefaciens*.
7. Rice embryo is infected with the *Agrobacterium tumefaciens* bacteria and the plasmids carrying the gene will be transferred into the plant cells.
8. The rice embryo will grow up into a mature plant that produces rice grains with high vitamin content

**(Max 6)**

(c) Discuss the pros and cons of genetically-modified crop plants.

[8]

1. Genetic engineering on plants can enhance crop yields.
2. Genetic engineering may permit crops to grow outside their usual location / season so that people have more food.
3. Genetically modified crop plants can also be enhanced with a certain nutritional content (e.g. Golden rice) so that people are better fed/OWTTE.
4. Genetically modified crop plants can be more pest-resistant (e.g BT corn),
5. and this will lower cost as pesticide usage will be reduced/avoided.
6. As pesticide usage is reduced/avoided, there will be less damage/pollution to the environment.
7. Genetically modified crop plants can be more drought-resistant, and this will increase crop yield for farmers.
8. Drought-resistant crops can also help farmers avoid installing costly irrigation systems to ensure sufficient water is provided.
9. As farmers' cost is reduced, their profits increase / consumer cost may also reduce.
10. Shelf-life of crops can be increased
11. The introduced gene(s) may be transferred by pollen to wild relatives whose hybrid offspring will become more invasive / become 'superweeds'.
12. This may lead to additional cost for the removal of such 'superweeds'.
13. 'Superweeds' may also reduce plant biodiversity by out-competing natural plants
14. e.g. Flavor Savr gene example must be given
15. The introduced gene(s) may be transferred by pollen to unmodified plants growing on a farm with 'organic' certification.
16. The modified plants will be a direct hazard to humans, domestic animals or other beneficial animals by being toxic.  
OR  
For instance, the herbicides that can now be used on the crop will itself leave toxic residues in the crop, which will be toxic to humans who consume them.  
e.g. Monarch Butterflies affected by BT.
17. The toxic residues may also affect and damage the environment by causing pollution to the surroundings.
18. The modified plants will be a direct hazard to humans, domestic animals or other beneficial animals by being producing allergies upon consumption.
19. Farmers may have to pay royalties to sow the crops.  
OR  
Some companies have extended their patents on genetically engineered seeds to prevent farmers from re-sowing the seed from these genetically engineered crops.
20. The cost that farmers have to pay for the royalties may be passed on the consumers who may have to pay higher prices for them.
21. Genetically identical and would be vulnerable to disease.

(Max 8, minimum 4 pros and 4 cons)

**END OF PAPER**